

Gait Phase Classification using Deep Neural Networks with Weighted Cross-Entropy Loss for handling Imbalanced Classes

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Abstract—Accurate gait-phase segmentation is essential to ensure stable, safe, and adaptive responses to biomechanical variability among individuals. However, despite advances in biomechanical instrumentation, traditional gait analysis systems remain costly and dependent on controlled environments, limiting their applicability in real-world and resource-constrained settings. This paper investigates self-attention-based Deep Neural Network (DNN) architectures for gait-phase classification using synchronized IMU signals collected from multiple lower-limb positions. Results show that self-attention-based DNNs achieve state-of-the-art performance with full sensor availability, while ensemble Machine Learning (ML) methods remain competitive under reduced-sensor configurations.

I. INTRODUCTION

Gait analysis is essential in assistive robotics and rehabilitation, as conditions such as Parkinson’s disease [1] and Alzheimer’s disease [2] manifest in distinct gait patterns characterized by slower pace and reduced stride amplitude. Its primary role is to provide precise measurements that differentiate normal from abnormal gait [3]. Traditionally performed with camera-based systems for patient monitoring and treatment selection [4], such methods often suffer from subjectivity and limited portability. Conventional gait-phase detection techniques further require frequent recalibration, as variations in anthropometry and walking style can degrade performance.

Artificial Intelligence (AI), particularly Machine Learning (ML) and Deep Learning (DL), addresses the limitations by learning complex patterns directly from multi-sensor IMU data, enabling consistent, adaptive, and objective gait-phase classification across diverse users and scenarios. This paper investigates three fully connected, self-attention-based Deep Neural Network (DNN) architectures (Models A, B, and C) alongside four established ML algorithms - Random Forest, XGBoost, Support Vector Machines (SVM), and Naïve Bayes - for classifying gait phases from wearable IMU data. A parallel evaluation of the DNN and ML pipelines is conducted under identical experimental conditions using

synchronized IMU datasets from multiple lower-limb positions, collected in [5], with class-weighted loss functions applied to mitigate class imbalance. Performance is assessed using accuracy, precision, recall, and F1-Score, supported by confusion matrix analysis. The goal is to determine whether self-attention-based DNN architectures can achieve superior classification performance while retaining robustness under different sensor configurations and across users.

II. RELATED WORK

A. Previous work

In our earlier papers [5], [6], the human gait cycle was segmented into five sub-phases: loading response, mid stance, terminal stance, initial swing, and terminal swing. Transitions between these phases were associated with characteristic gait events: Heel Strike (Hs), marking the first foot-ground contact; Toe Strike (TS), corresponding to full foot-ground contact; Heel Off (HO) and Toe Off (TO), indicating loss of heel and toe contact, respectively; and, Mid Swing (MSw), when the balancing and support legs are adjacent.

In [6], we represented the phase sequence using a Finite State Machine (FSM), where each state corresponds to a gait sub-phase. Phase transitions were modeled as a discrete-time Markov chain, with the sequence of phases governed by transition probabilities. In [5], we employed four IMUs, mounted on the thigh, knee, shank, and heel, to provide inputs to a Kalman filter estimator, which identified the current gait phase as well as the angular position and velocity of the knee joint. This information enabled the controller to minimize human–robot interaction forces during phase transitions.

However, these studies did not address scalability to more constrained scenarios or evaluate model robustness when individual sensors were removed or failed. This limitation motivates the present work, which extends our previous research by investigating gait phase identification under reduced-sensor configurations. Specifically, we evaluate both deep learning and classical machine learning methods when the heel sensor is removed, leaving only three IMUs. This scenario is designed to assess the feasibility of lightweight models for practical deployment in rehabilitation and assistive technologies, where robust and cost-effective solutions are essential.

B. Machine Learning applied to human gait identification

Recent studies have applied a variety of machine learning strategies to human gait identification, ranging from

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